

Complete documentation of all user roles, features, API connections, user flows, and data flows.

- 1. [User Roles Overview](#1-user-roles-overview)
- 2. [User Accounts & Features](#2-user-accounts--features)
- 3. [API Architecture](#3-api-architecture)
- 4. [User Flows](#4-user-flows)
- 5. [Data Flows](#5-data-flows)
- 6. [Database Schema](#6-database-schema)
- 7. [Role Permissions Matrix](#7-role-permissions-matrix)

I Role I	Code I	Description I
I-----I	I-----I	I-----I
I Super Admin I	I `superadmin` I	I Full platform control, user management, inventory I
I Admin I	I `admin` I	I Platform operations, approvals I
I Doctor I	I `doctor` I	I Treatment planning, patient management I
I Nurse I	I `nurse` I	I Treatment execution, infusion preparation I
I Clinic I	I `clinic` I	I Partner clinic operations, bookings I
I Patient I	I `patient` I	I End user, health quiz, booking I

****Access:**** `/dashboard/admin`
****Role Codes:**** `superadmin`, `admin`

Features:

I Feature I	Module I	Description I
I-----I	I-----I	I-----I
I Dashboard I	I Core I	I Platform stats, revenue, user counts I
I User Management I	I Manage I	I CRUD all users (patients, doctors, nurses, clinics) I
I Inventory I	I Inventory I	I Products, batches, drip builder, availability I
I AI Studio I	I Studio I	I Configure AI models I
I Approvals I	I Dashboard I	I Review pending requests I

Demo Credentials:
...

Email: admin@nutridrip.com

Password: admin123

...

Data Model:

``typescript

```
{
  id: string;
  email: string;
  name: string;
  phone?: string;
  role: "superadmin" | "admin";
  permissions: string[]; // JSON array
  status: "active" | "inactive";
  createdAt: Date;
  updatedAt: Date;
}
```

...

****Access:**** `/dashboard/doctor`

****Role Code:**** `doctor`

Features:

| Feature | Module | Description |

|-----|-----|-----|

| Dashboard | Core | Patient stats, pending quizzes |

| Health Quizzes | Quiz | Review patient vitality assessments |

| Treatment Plans | TxBuilder | Create multi-week IV therapy plans |

| Sessions | TxBuilder | Configure drip components, doses, routes |

| Share Plans | TxBuilder | Send plans to nurses for execution |

Demo Credentials:

...

Email: dr.sarah@nutridrip.com

Password: doctor123

...

Data Model:

``typescript

```
{
  id: string;
  email: string;
  name: string;
  phone?: string;
  role: "doctor";
  specialization?: string;
  licenseNo?: string;
  createdAt: Date;
}
```

...

Treatment Plan Structure:

``typescript

```

{
  id: string;
  patientName: string;
  patientAge: string;
  diagnosis: string;
  startDate: string;
  totalWeeks: number;
  weeks: [
    {
      weekNum: number;
      sessions: [
        {
          id: string;
          date: string;
          dripName: string;
          components: [
            {
              id: string;
              name: string;
              dose: string;
              unit: string;
              route: "IV Drip in NS" | "IV Drip in RL" | "IV Push/Bolus" | "IM Injection" | "Add to Drip Bag" | "Oral";
              carrier: string;
            }
          ];
        }
      ];
      sessionNotes: string;
    }
  ];
  sharedWithNurse: boolean;
  doctorName: string;
}

```

Access: `/dashboard/nurse`

Role Code: `nurse`

Features:

| Feature | Module | Description |

|-----|-----|-----|

| Dashboard | Core | Order stats, today's tasks |

| Infusion Orders | Orders | View assigned treatment plans |

| Infusion Checklist | Checklist | Prepare drips step-by-step |

| Complete Sessions | Orders | Mark treatments done |

Demo Credentials:

...

Email: nurse.emma@nutridrip.com

Password: nurse123

...

Data Model:

```typescript

```
{
 id: string;
 email: string;
 name: string;
 phone?: string;
 role: "nurse";
 licenseNo?: string;
 assignedTo?: string; // clinic ID
 createdAt: Date;
}
```

...

#### #### Infusion Order Structure:

```typescript

```
{
  id: string;
  patientName: string;
  dripName: string;
  scheduledAt: string;
  status: "pending" | "in-progress" | "completed" | "cancelled";
  location: "home" | "clinic" | "office" | "hotel";
  notes?: string;
  nurseAssigned: string | null;
  checklist: string[]; // preparation steps
}
```

...

****Access:**** `/dashboard/clinic`

****Role Code:**** `clinic`

Features:

| Feature | Module | Description |

|-----|-----|-----|

| Dashboard | Core | Orders, revenue, bookings stats |

| Orders | Orders | View clinic orders |

| Bookings | Bookings | View patient appointments |

| Profile | Profile | View clinic info |

Demo Credentials:

...

Email: clinic@healthfirst.com

Password: clinic123

...

Data Model:

```typescript

```
{
 id: string;
 email: string;
 name: string;
 phone?: string;
 role: "clinic";
 address?: string;
 location?: string;
 status: "active" | "inactive" | "pending";
 partnersSince?: Date;
 monthlyVolume?: number;
}
```

#### Clinic Order Structure:

```
``typescript
{
 id: string;
 orderedAt: string;
 items: string; // e.g., "Velocity x 5, Cognitas x 6"
 amount: number;
 status: "pending" | "processing" | "dispatched" | "delivered" | "cancelled";
 scheduledDelivery?: Date;
}
```

---

```
Access: `/dashboard/patient`
Role Code: `patient`
```

#### Features:

| Feature      | Module  | Description                       |
|--------------|---------|-----------------------------------|
| Dashboard    | Core    | Upcoming sessions, vitality score |
| Health Quiz  | Quiz    | Complete vitality assessment      |
| Book Session | Book    | Schedule appointment              |
| Lab Reports  | Lab     | Upload medical documents          |
| Profile      | Profile | Update health info                |

#### Demo Credentials:

```
...
Email: patient@example.com
Password: patient123
...
```

#### Data Model:

```
``typescript
{
 id: string;
 email: string;
 name: string;
 phone?: string;
```

```

 role: "patient";
 dob?: string;
 bloodGroup?: string;
 address?: string;
 emergencyContact?: string;
 allergies?: string;
 chronicConditions?: string;
 currentMedications?: string;
 surgeries?: string;
 familyHistory?: string;
 lifestyleNotes?: string;
 vitalityScore?: number;
 createdAt: Date;
 }
 ...

```

#### #### Health Quiz Structure:

```

````typescript
{
  id: string;
  patientId: string;
  responses: {
    energy: number; // 1-10
    sleep: number; // 1-10
    stress: number; // 1-10
    nutrition: number; // 1-10
    exercise: number; // 1-10
    hydration: number; // 1-10
  };
  vitalityScore: number; // calculated average
  recommendations: string[];
  createdAt: Date;
}
...

```

Lab Report Structure:

```

````typescript
{
 id: string;
 fileName: string;
 uploadedAt: Date;
 sizeBytes: number;
 category: string;
 notes?: string;
 url?: string;
}
...

```

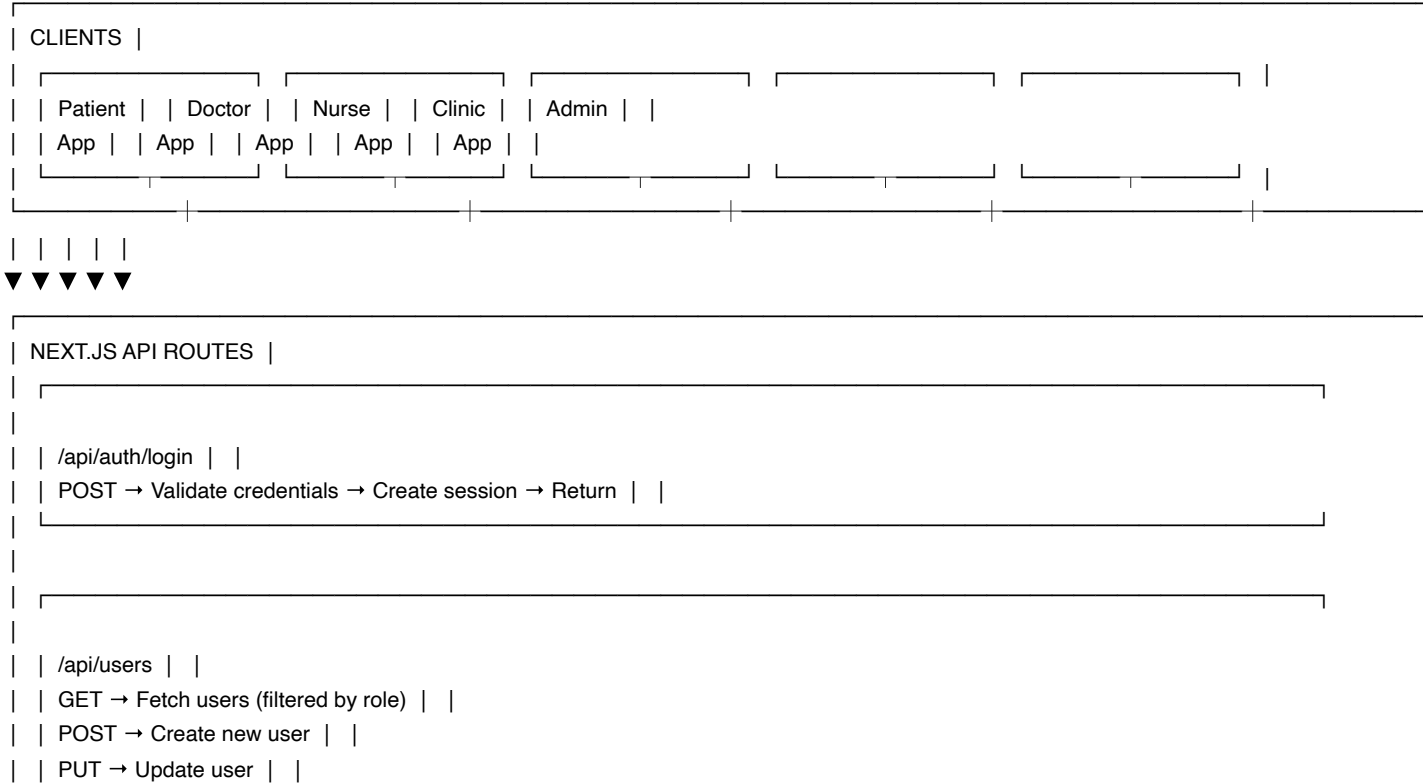
---

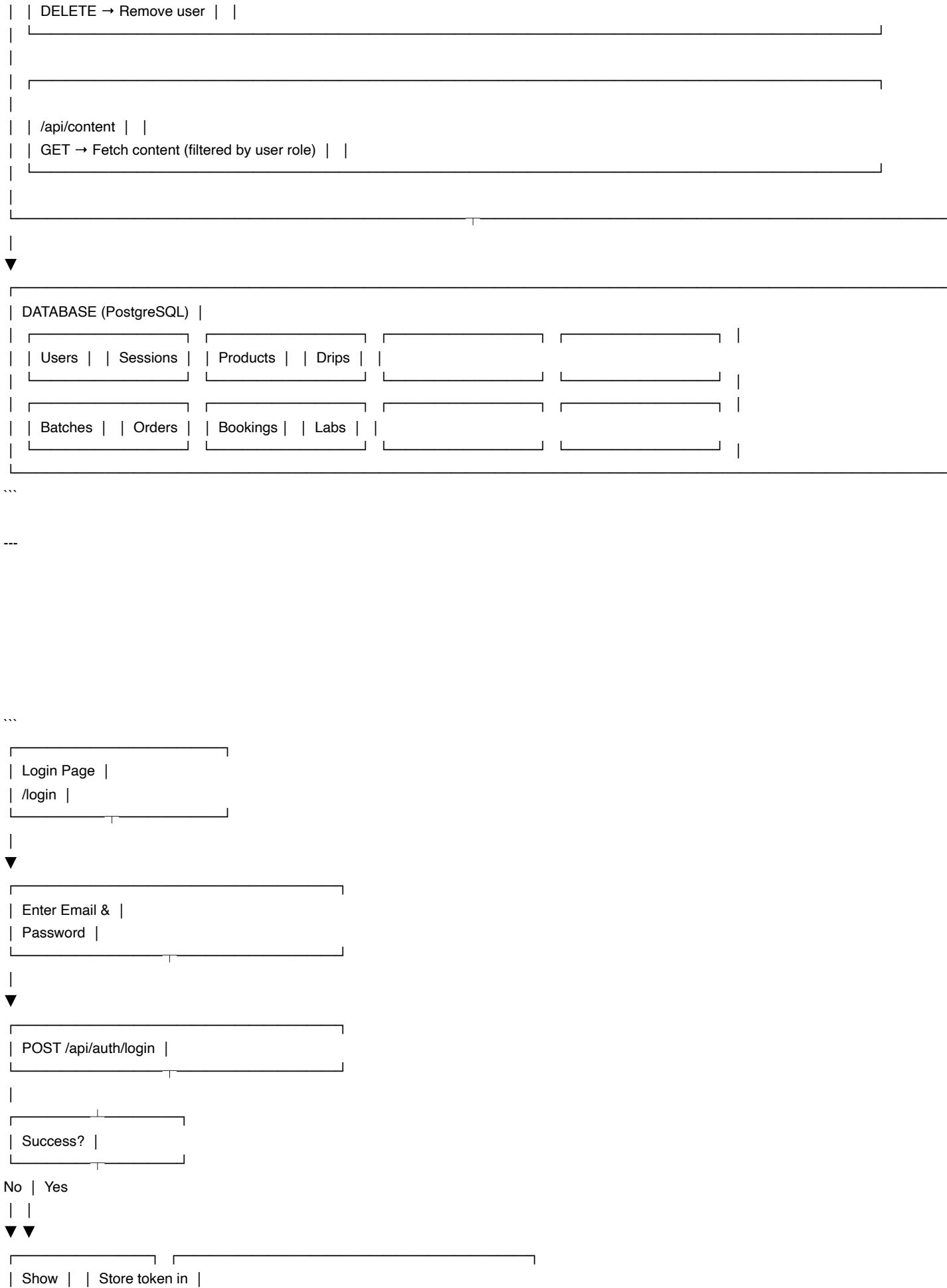
| Method | Endpoint        | Description    | Auth Required | Roles              |
|--------|-----------------|----------------|---------------|--------------------|
| POST   | /api/auth/login | User login     | No            | All                |
| GET    | /api/users      | List users     | Yes           | Admin, Super Admin |
| GET    | /api/users?id=  | Get user by ID | Yes           | Admin, Super Admin |
| POST   | /api/users      | Create user    | Yes           | Super Admin        |
| PUT    | /api/users?id=  | Update user    | Yes           | Super Admin        |
| DELETE | /api/users      | Delete user    | Yes           | Super Admin        |
| GET    | /api/content    | Get content    | Yes           | All (filtered)     |

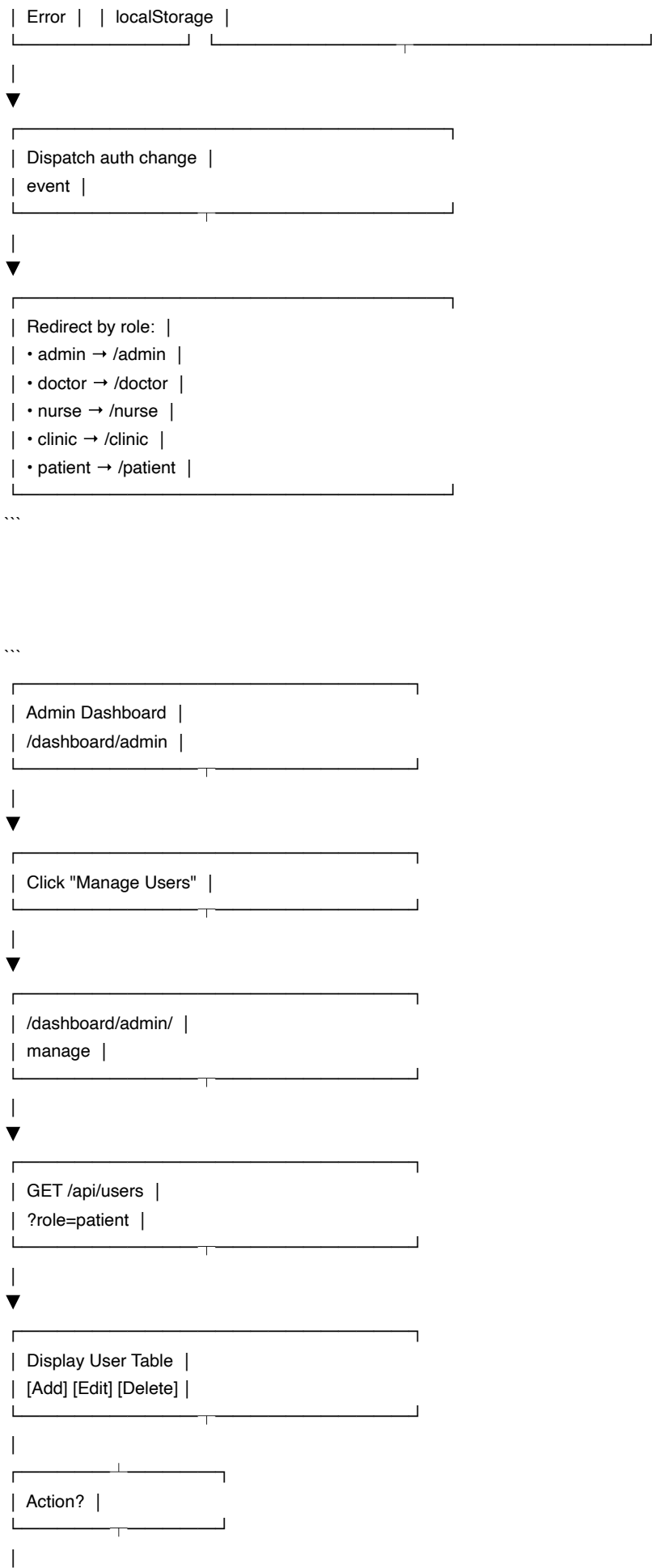
```
````typescript
// Success Response
{
  success: true,
  data: T | T[],
  message?: string
}

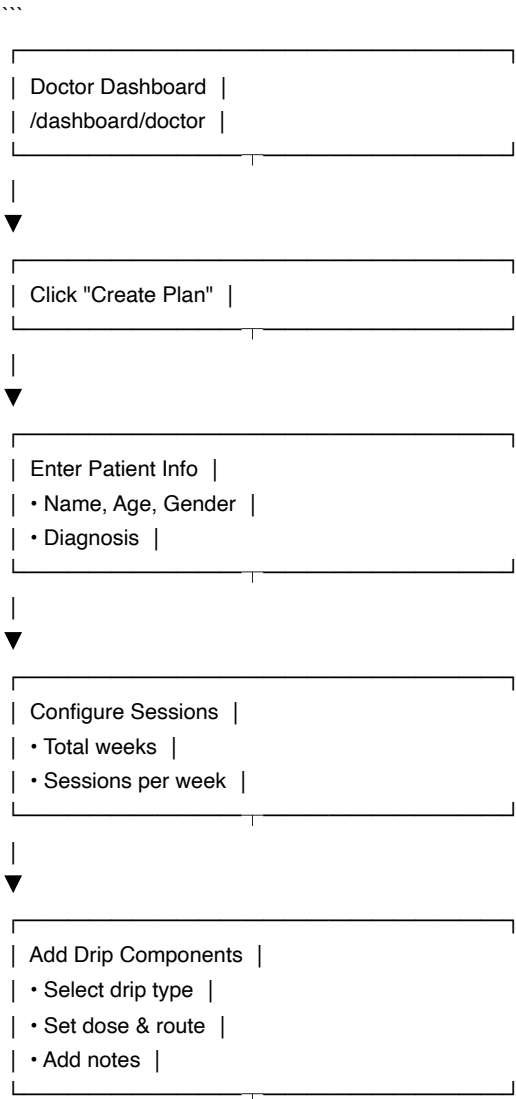
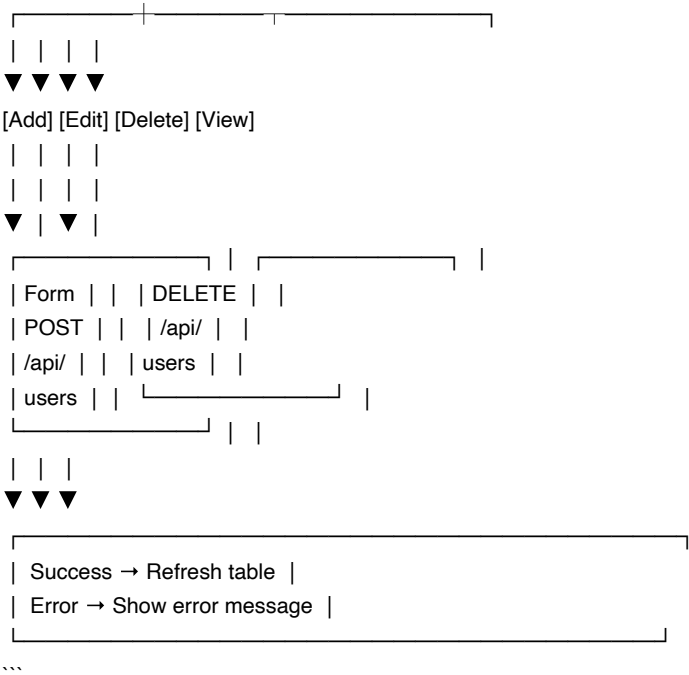
// Error Response
{
  success: false,
  error: string,
  code?: string
}
...
...
...

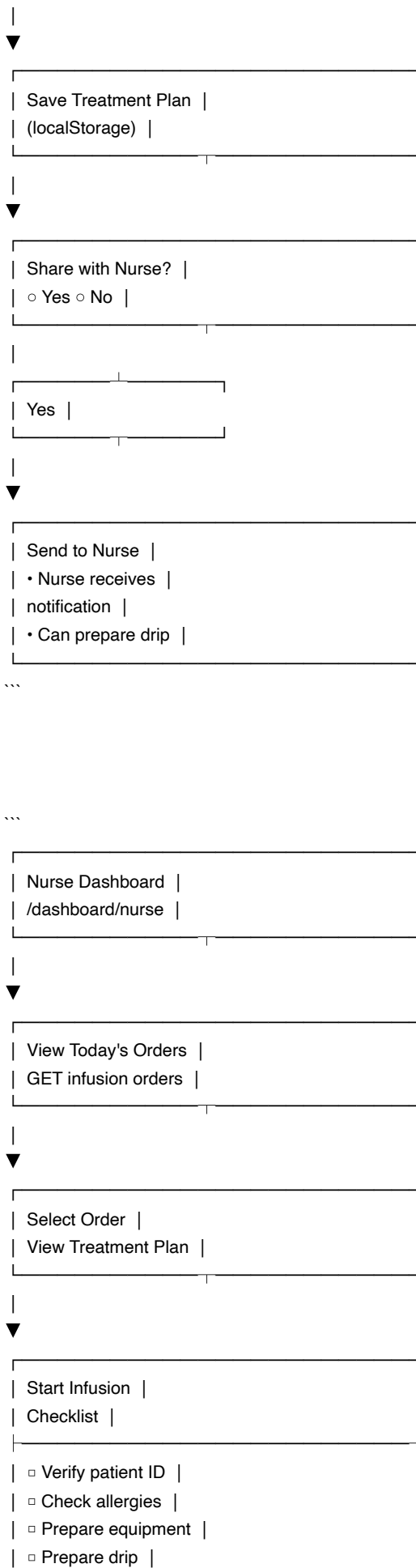
```

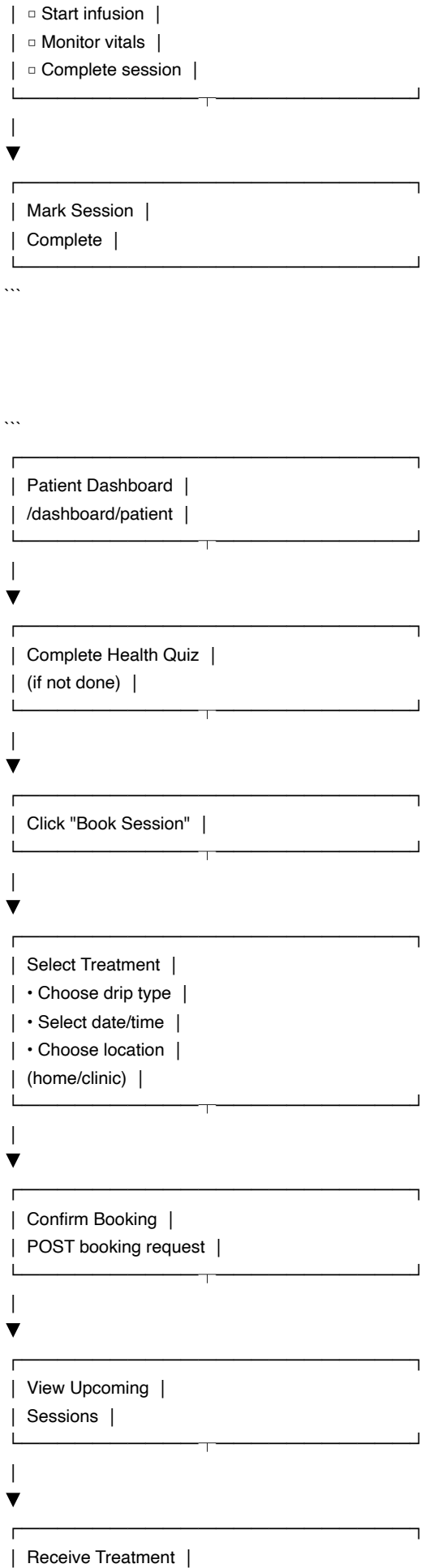


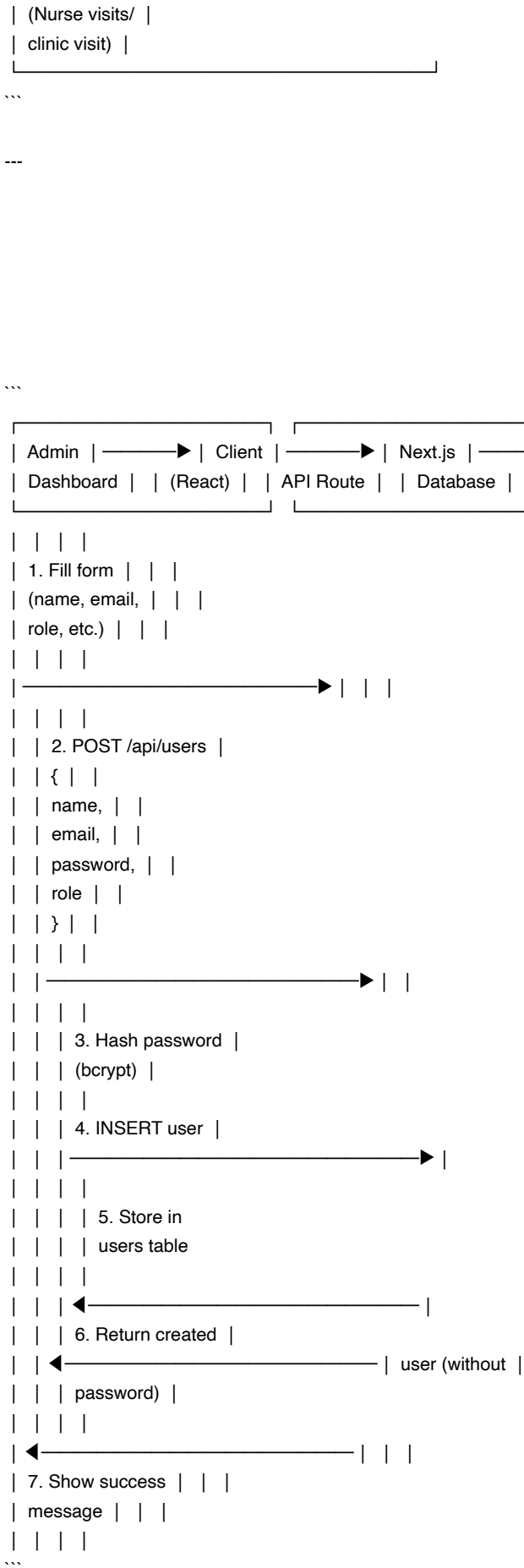




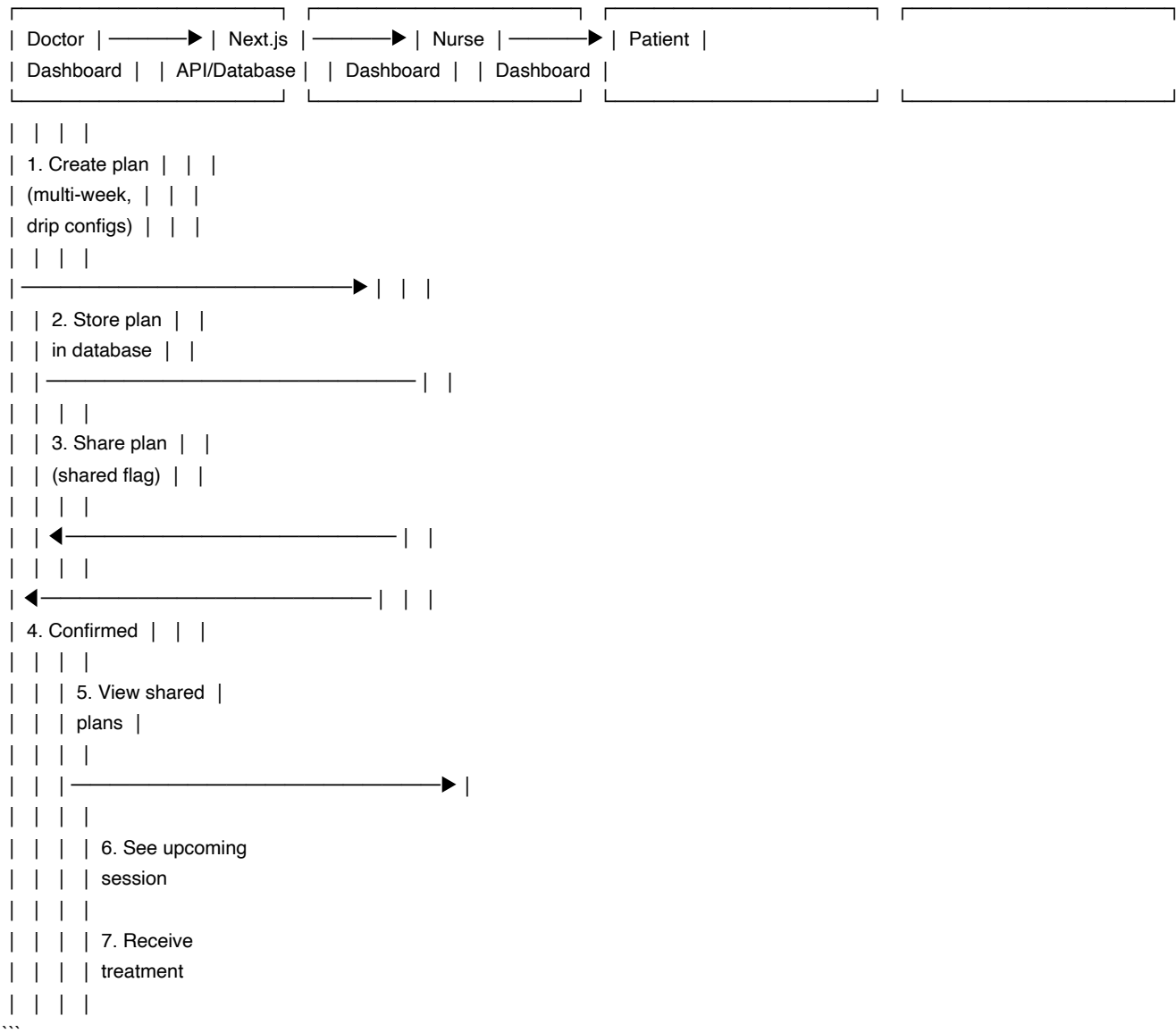




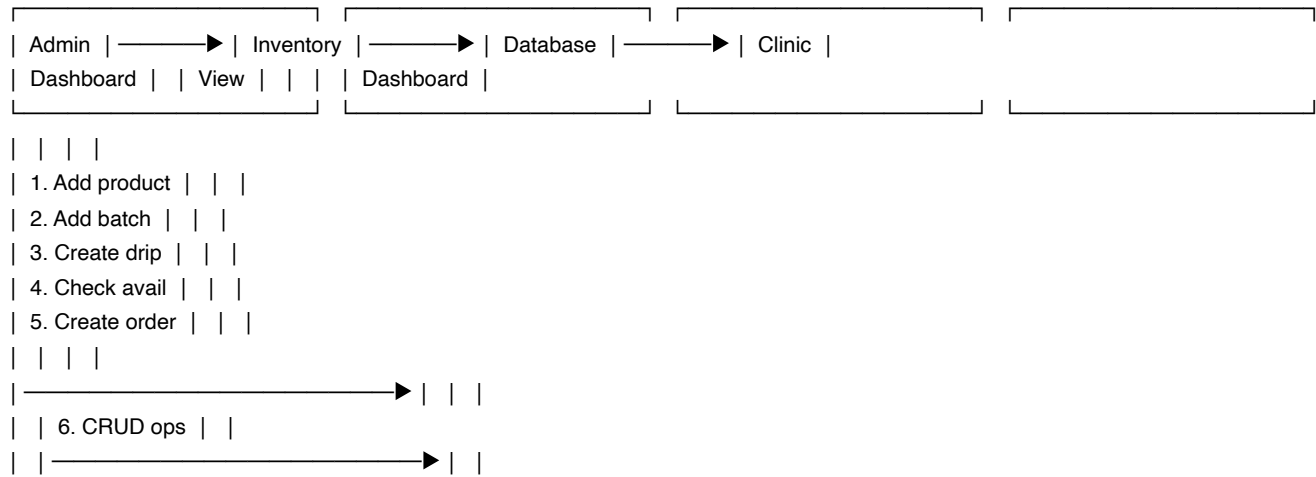




...



...



```

| | | 7. Update tables: | | |
| | | • products |
| | | • batches |
| | | • drips |
| | | • orders |
| | | |
| | | ← | | |
| 8. View updated | | |
| inventory | | |
| | | ← |
| | | 9. Stock update |
| | | notification |
| | | |
| | | → |
| | | 10. View order
| | | status
| | | |
...

```

```
---
```

```

```sql
-- Users table (all roles)
CREATE TABLE users (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
email VARCHAR(255) UNIQUE NOT NULL,
password VARCHAR(255) NOT NULL, -- bcrypt hashed
name VARCHAR(255) NOT NULL,
phone VARCHAR(50),
role VARCHAR(50) NOT NULL CHECK (role IN ('superadmin', 'admin', 'doctor', 'nurse', 'clinic', 'patient')),
permissions JSONB DEFAULT '[]',
status VARCHAR(20) DEFAULT 'active',
created_at TIMESTAMP DEFAULT NOW(),
updated_at TIMESTAMP DEFAULT NOW()
);

-- Sessions table
CREATE TABLE sessions (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
user_id UUID REFERENCES users(id),
token VARCHAR(255) UNIQUE NOT NULL,
expires_at TIMESTAMP NOT NULL,
created_at TIMESTAMP DEFAULT NOW()
);

-- Products (inventory)
CREATE TABLE products (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
name VARCHAR(255) NOT NULL,
hsn_code VARCHAR(50),

```

```
category VARCHAR(50), -- DRUG, FLUID, CONSUMABLE, PREMED
molecule VARCHAR(255),
reorder_level INTEGER DEFAULT 0,
is_multidose BOOLEAN DEFAULT false,
is_active BOOLEAN DEFAULT true,
created_at TIMESTAMP DEFAULT NOW()
);
```

```
-- Batches (inventory lots)
CREATE TABLE batches (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
product_id UUID REFERENCES products(id),
brand_name VARCHAR(255),
manufacturer VARCHAR(255),
batch_number VARCHAR(100),
expiry_date DATE NOT NULL,
content_value DECIMAL(10,2),
content_unit VARCHAR(50),
unit_form VARCHAR(100),
qty_on_hand INTEGER DEFAULT 0,
is_active BOOLEAN DEFAULT true,
created_at TIMESTAMP DEFAULT NOW()
);
```

```
-- Drips (formulas)
CREATE TABLE drips (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
name VARCHAR(255) NOT NULL,
description TEXT,
infusion_notes TEXT,
with_kit BOOLEAN DEFAULT false,
is_active BOOLEAN DEFAULT true,
created_by UUID REFERENCES users(id),
created_at TIMESTAMP DEFAULT NOW(),
updated_at TIMESTAMP DEFAULT NOW()
);
```

```
-- Drip ingredients
CREATE TABLE drip_ingredients (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
drip_id UUID REFERENCES drips(id) ON DELETE CASCADE,
product_id UUID REFERENCES products(id),
dose DECIMAL(10,2) NOT NULL,
unit VARCHAR(50) NOT NULL,
role VARCHAR(50) NOT NULL, -- ACTIVE, FLUID, PREMED, ADDITIVE
created_at TIMESTAMP DEFAULT NOW()
);
```

```
-- Orders
CREATE TABLE orders (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
ordered_by UUID REFERENCES users(id),
status VARCHAR(50) DEFAULT 'draft',
amount DECIMAL(12,2) DEFAULT 0,
scheduled_delivery DATE,
notes TEXT,
```

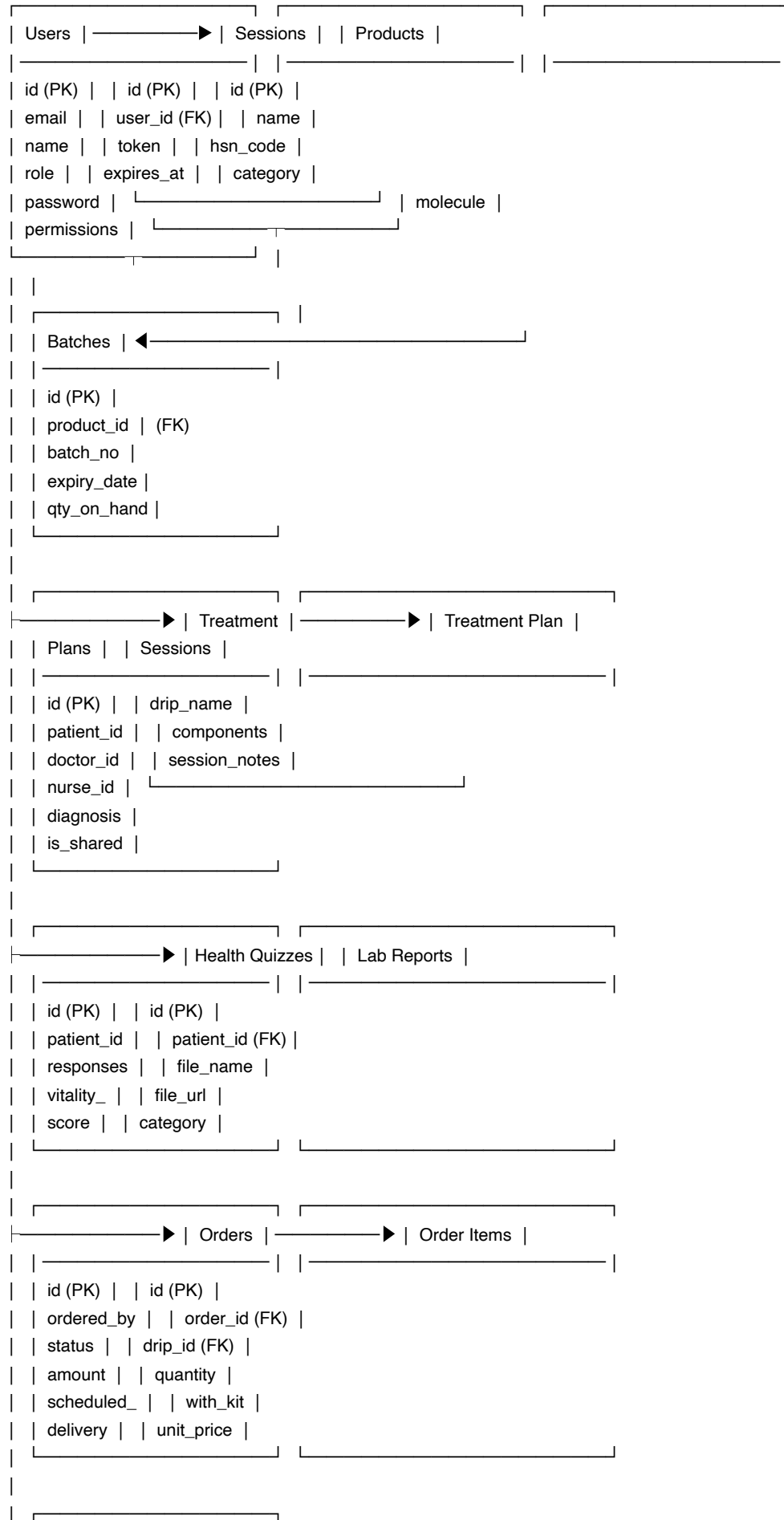
```
created_at TIMESTAMP DEFAULT NOW(),
updated_at TIMESTAMP DEFAULT NOW()
);

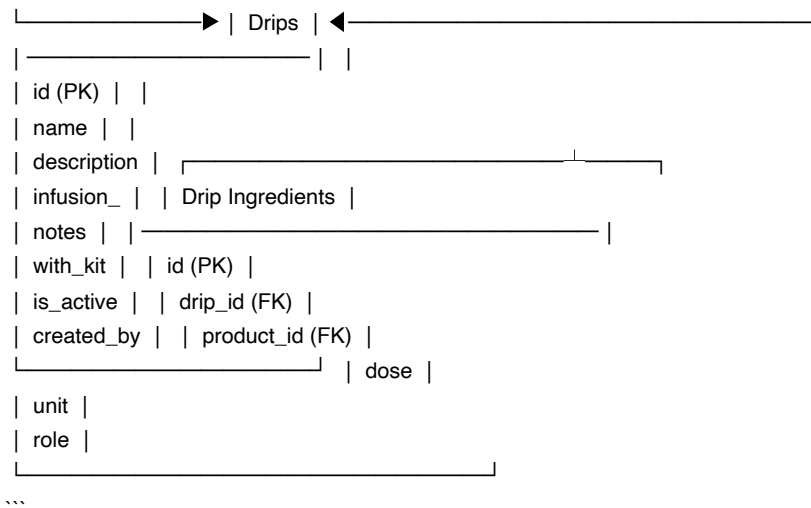
-- Order items
CREATE TABLE order_items (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
order_id UUID REFERENCES orders(id) ON DELETE CASCADE,
drip_id UUID REFERENCES drips(id),
quantity INTEGER NOT NULL,
with_kit BOOLEAN DEFAULT false,
unit_price DECIMAL(10,2),
created_at TIMESTAMP DEFAULT NOW()
);

-- Treatment plans
CREATE TABLE treatment_plans (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
patient_id UUID REFERENCES users(id),
doctor_id UUID REFERENCES users(id),
nurse_id UUID REFERENCES users(id),
diagnosis TEXT,
start_date DATE,
total_weeks INTEGER,
is_shared BOOLEAN DEFAULT false,
status VARCHAR(50) DEFAULT 'draft',
created_at TIMESTAMP DEFAULT NOW(),
updated_at TIMESTAMP DEFAULT NOW()
);

-- Health quizzes
CREATE TABLE health_quizzes (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
patient_id UUID REFERENCES users(id),
responses JSONB NOT NULL,
vitality_score INTEGER,
recommendations JSONB,
created_at TIMESTAMP DEFAULT NOW()
);

-- Lab reports
CREATE TABLE lab_reports (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
patient_id UUID REFERENCES users(id),
file_name VARCHAR(255) NOT NULL,
file_url TEXT,
file_size INTEGER,
category VARCHAR(100),
notes TEXT,
uploaded_at TIMESTAMP DEFAULT NOW()
);
...
```





...

| Feature | Super Admin | Admin | Doctor | Nurse | Clinic | Patient |

-----|:-----|:-----|:-----|:-----|:-----|:-----|:-----|

| **\*\*Authentication\*\*** |

| Login | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

| Logout | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

```
| **Users** |
```

| View all users | ✓ | ✓ | - | - | - | - |

| Create user | ✓ | - | - | - | - | - |

| Edit user | ✓ | - | - | - | - | - |

| Delete user | ✓ | - | - | - | - | - |

| **\*\*Inventory\*\*** |

| View products | ✓ | ✓ | - | - | - | - |

| Manage products | ✓ | - | - | - | - | - |

| View batches | ✓ | ✓ | - | - | - | - |

|                |   |   |   |   |   |   |
|----------------|---|---|---|---|---|---|
| Manage batches | ✓ | - | - | - | - | - |
|----------------|---|---|---|---|---|---|

| Build drips | ✓ | - | - | - | - | - |

| Check availability | ✓ | ✓ | - | - | - | - |

| **\*\*Orders\*\*** |

| View orders | ✓ | ✓ | ✓ | ✓ | ✓ | - |

| Create order | ✓ | ✓ | - | - | ✓ | - |

| Update order | ✓ | ✓ | - | - | - | - |

| \*\*Treatment\*\* |

| View treatment plans | ✓ | ✓ | ✓ | ✓ | - | ✓ |

| Create treatment plan | ✓ | - | ✓ | - | - | - |

| Share with nurse | ✓ | - | ✓ | - | - | - |

| Prepare infusion | ✓ | - | - | ✓ | - | - |

| Complete session | ✓ | - | - | ✓ | - | - |

| \*\*Health\*\* |

| Take health quiz | - | - | - | - | - | ✓ |

| Review quizzes | ✓ | ✓ | ✓ | - | - | - |

| Upload lab reports | - | - | - | - | - | ✓ |

| View lab reports | ✓ | ✓ | ✓ | - | - | ✓ |

```
| **Bookings** |
```

| Book session | - | - | - | - | - | ✓ |

| View bookings | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

| Manage bookings | ✓ | ✓ | - | - | - | - |  
| **Clinic** |  
| View clinic profile | ✓ | ✓ | - | - | - | ✓ | - |  
| Update clinic info | ✓ | - | - | - | - | ✓ | - |  
| **AI** |  
| Configure AI models | ✓ | - | - | - | - | - | - |  
| View AI settings | ✓ | - | - | - | - | - | - |  
| **Approvals** |  
| View approvals | ✓ | ✓ | ✓ | ✓ | - | - | - |  
| Approve/Reject | ✓ | ✓ | ✓ | ✓ | - | - | - |

---

\*Document Version: 1.0\*  
\*Last Updated: August 2026\*